

FULL STACK WEB DEVELOPMENT - SOURCE CODE

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Question 3: Express.js Student Server

File: server.js

```
// server.js
// Basic Express.js Student Server
// Routes: / (home), /students (student list), /about (app information)

// Import the Express module (installed using: npm install express)
const express = require("express");

// Create the Express application
const app = express();
// Port number on which the server will listen
const PORT = 3000;

// Pretty-print JSON responses (2 spaces) so they are easy to read
app.set("json spaces", 2);

// Sample student data (at least five students)
const students = [
  { id: 1, name: "Ananya Reddy", department: "CSE", year: 3 },
  { id: 2, name: "Rahul Sharma", department: "CSE", year: 3 },
  { id: 3, name: "Priya Nair", department: "IT", year: 3 },
  { id: 4, name: "Arjun Verma", department: "ECE", year: 3 },
  { id: 5, name: "Kiran Kumar", department: "CSE", year: 3 },
  { id: 6, name: "Sneha Patel", department: "IT", year: 3 }
];

// GET / : home route - shows a welcome page with links to other routes
// (req = incoming request, res = response sent back to the browser)
app.get("/", (req, res) => {
  res.send(`
    <h1>Welcome to the Express.js Student Server</h1>
    <p>Available routes:</p>
    <ul>
      <li><a href="/students">/students</a> - list of students</li>
      <li><a href="/about">/about</a> - about this application</li>
    </ul>
  `);
});

// GET /students : returns the list of students in JSON format
app.get("/students", (req, res) => {
  res.json(students);
});

// GET /about : displays information about the application
app.get("/about", (req, res) => {
  res.send(`
    <h1>About This Application</h1>
    <p><strong>Application:</strong> Express.js Student Server</p>
    <p><strong>Built with:</strong> Node.js and Express.js</p>
    <p><strong>Purpose:</strong> Demonstrates server creation, routing
    and HTTP GET requests.</p>
    <p><strong>Version:</strong> 1.0</p>
  `);
});

// Middleware placed after all routes: runs for any URL that did not match
// above, so unknown pages get a 404 (Not Found) response
app.use((req, res) => {
  res.status(404).send("<h1>404 - Page Not Found</h1>");
});

// Start the server and print a message in the terminal
app.listen(PORT, () => {
  console.log(`Server running at http://localhost:${PORT}`);
});
```

File: package.json

```
{
  "name": "express-student-server",
  "version": "1.0.0",
  "description": "Basic Express.js server with /, /students and /about routes",
  "main": "server.js",
  "scripts": {
    "start": "node server.js"
  },
  "author": "Kottana Roshini",
  "license": "ISC",
  "dependencies": {
    "express": "^4.21.2"
  }
}
```